

Workers wake up the manager.

```
void worker_A ( ) {
    int heart = 0;
    while(true) { semWait(m);
    make_candy ( );
    heart++; semWait (box);
    put_in_box ( );
    semSignal (box);
    if(heart == 6) { heart = 0;
    semSignal (d); } } }
```

```
void manager ( ) {
    while(true){
    semWait (d);
    semWait (d);
    provide_chocolate ( );
    provide_box ( );
    semSignal (m);
    semSignal (m); } }
```

```
void storekeeper ( ) {
    while(true){
    semWait (s);
    /*give_almonds_to_worker_B*/;
    semSignal (b);
    } }
```

```
void worker_B ( ) {
    int almond = 0;
    while(true) { semWait(m);
    if(out_of_almonds) {semSignal (s);
    /*wake_up_storekeeper */;
    semWait (b) }
    make_candy();
    almond++; semWait (box);
    put_in_box ( ); semSignal (box);
    if(almond == 6) { almond = 0;
    semSignal (d); } } }
```

```
semaphore m = 0;
semaphore box = 1;
semaphore s = 0; semaphore b = 0;
semaphore d = 2;
void main ( ) {
    parbegin(manager, storekeeper, worker_A, worker_B);
    }
```

This is slide# 10 of lecture 18.

Corrected code of worker A:

```
void worker_A ( ) {
    int heart = 0;
    while(true) {
    if (heart == 0) semWait(m);
    make_candy ( );
    ...
```

The rest of the code is unchanged.

Corrected code of worker B:

```
void worker_B ( ) {
    int almond = 0;
    while(true) {
    if (almond == 0) semWait(m);
    if(out_of_almonds) {semSignal (s);
    ...
```

The rest of the code is unchanged.